VOLUME II: RIGID-BODY & MULTIBODY DYNAMICS OF THE SHOVEL SYSTEM

CHAPTER 6: Top-Down Kinematic & Topological Decomposition of the Electric Rope Shovel

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Target Machines: Machine A (P&H 2100BL Benchmark) & Machine B (TZ WK-20 Deployment)

Pedagogical Level: Postgraduate / Research Grade (B.Tech Mechanical Foundation Assumed)

NOTE

RESEARCH STUDY NOTEBOOK ARCHITECTURE

In Volume I (Chapters 1–5), we constructed the rigorous mathematical, variational, and observer foundations for nonlinear multibody systems. We now enter **Volume II**, where we deploy these analytical tools onto the complete physical architecture of the electric rope shovel.

To model a massive $800\,\mathrm{t}$ mining excavator accurately, an engineer cannot treat the machine as an abstract mathematical black box. You must understand how every pin, pinion, sheave, and wire rope constraints the motion of the dipper.

By the end of this chapter, you will be able to:

1. Decompose the complete industrial shovel into its five fundamental structural subsystems and construct its topological kinematic graph.

2. Define a unified, non-ambiguous tree of orthogonal coordinate frames from the ground world frame $\{W\}$ through the revolving frame $\{0\}$, boom $\{1\}$, saddle $\{3\}$, handle $\{4\}$, dipper $\{5\}$, and tooth tip $\{T\}$.

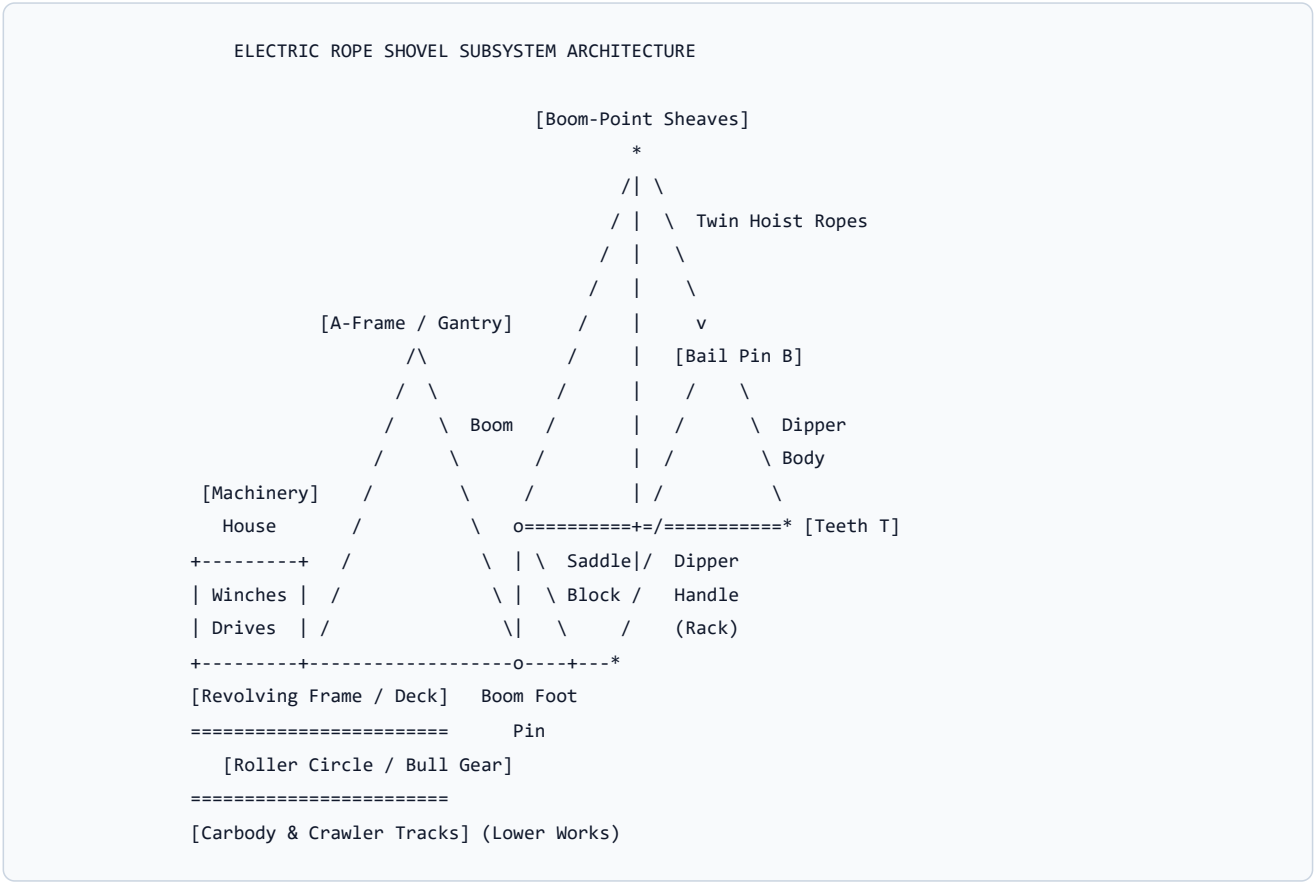
3. Formulate the non-linear **Closed Hoist Cable Loop** equation and derive analytical expressions for rope length $L_h(q_3, d_4)$ and hoist moment arm $h_h(q_3, d_4)$.

4. Synthesize the full 3D spatial forward kinematics incorporating the upper works slewing/swing angle θ_1.

5. Map the structural parameters and kinematic differences between the **P&H 2100BL benchmark** ($50\,\mathrm{t}$ class) and the **TZ WK-20 deployment shovel** ($55\,\mathrm{t}$ payload class).

6.1 Physical Overview: The Five Subsystems of an Electric Rope Shovel

An electric rope shovel is a heavy-duty, cyclic excavation machine engineered to dig, hoist, swing, and dump dense blasted rock, coal, or iron ore into off-highway haul trucks. The machine consists of five interconnected mechanical subsystems:



6.1.1 Subsystem 1: Lower Works (Carbody & Crawlers)

- **Carbody (Truck Frame):** A massive central alloy-steel casting that supports the entire weight of the upper machine ($350\,\mathrm{t} - 900\,\mathrm{t}$). It houses the central vertical gantry pin (kingpin) and propels hydraulic or mechanical travel gearing.
- **Crawler Assemblies:** Dual heavy crawler tracks provide low ground-bearing pressure ($150\,\mathrm{kPa} - 280\,\mathrm{kPa}$) across rough pit floors. During the active digging cycle, the propel brakes are locked, and the carbody acts as a **rigid inertial foundation**.
- **Roller Circle & Swing Rack:** A heavy circular forged path carrying 30–50 tapered alloy-steel rollers. Concentric with the roller circle is the internal-toothed or external-toothed **Bull Gear** ($3.5\,\mathrm{m} - 5.5\,\mathrm{m}$ diameter) with which the vertical swing motor pinions mesh.

6.1.2 Subsystem 2: Upper Works (Revolving Frame & Machinery House)

- **Revolving Deck:** A heavy-duty, torsionally rigid welded deck carrying all electrical transformers, variable frequency drives (VFDs), motor-generator sets, hoist winches, swing transmissions, and the air compressor system.
- **A-Frame Gantry:** A high triangular structural gantry mounted over the machinery house. It anchors the boom suspension bridge ropes, maintaining boom elevation under digging reaction loads.
- **Counterweight:** Ballast steel and slag blocks housed in the rear overhang of the deck to counterbalance the forward overturning moment of the extended dipper.

6.1.3 Subsystem 3: Boom Assembly

- **Boom Foot Pins:** Dual heavy alloy-steel pins O_1 anchoring the boom heel to the front deck.
- **Chords and Lacings:** Two wide-flange tubular or box-section chords inclined at fixed angle $\alpha_b \approx 45^\circ$, reinforced with cross-lacings to resist bending and torsional shock.
- **Shipper Shaft Assembly:** Located along the boom at pivot O_3 ($8.84\,\mathrm{m}$ from heel on P&H 2100BL, $11.20\,\mathrm{m}$ on WK-20). It houses the crowd pinion shaft, crowd drive motor/transmission, and saddle block trunnion bearings.
- **Boom-Point Sheaves:** Twin high-hardness cast alloy-steel sheaves mounted on anti-friction roller bearings at the boom tip, carrying the dual hoist ropes.

6.1.4 Subsystem 4: Crowd & Dipper Handle Mechanism

- **Saddle Block:** A heavy pivoting saddle casting mounted on the shipper shaft bronze bushings. It swivels freely in the vertical plane, guiding the handle through bronze or polymer wear plates.
- **Dipper Handle:** Dual parallel box-beam or tubular cylindrical steel arms ($10\,\mathrm{m} - 16\,\mathrm{m}$ length) with cast manganese alloy-steel gear racks welded along their bottom edges.
- **Crowd Pinion Mesh:** Heavy spur pinions on the shipper shaft mesh directly with the handle racks to push or retract the dipper.

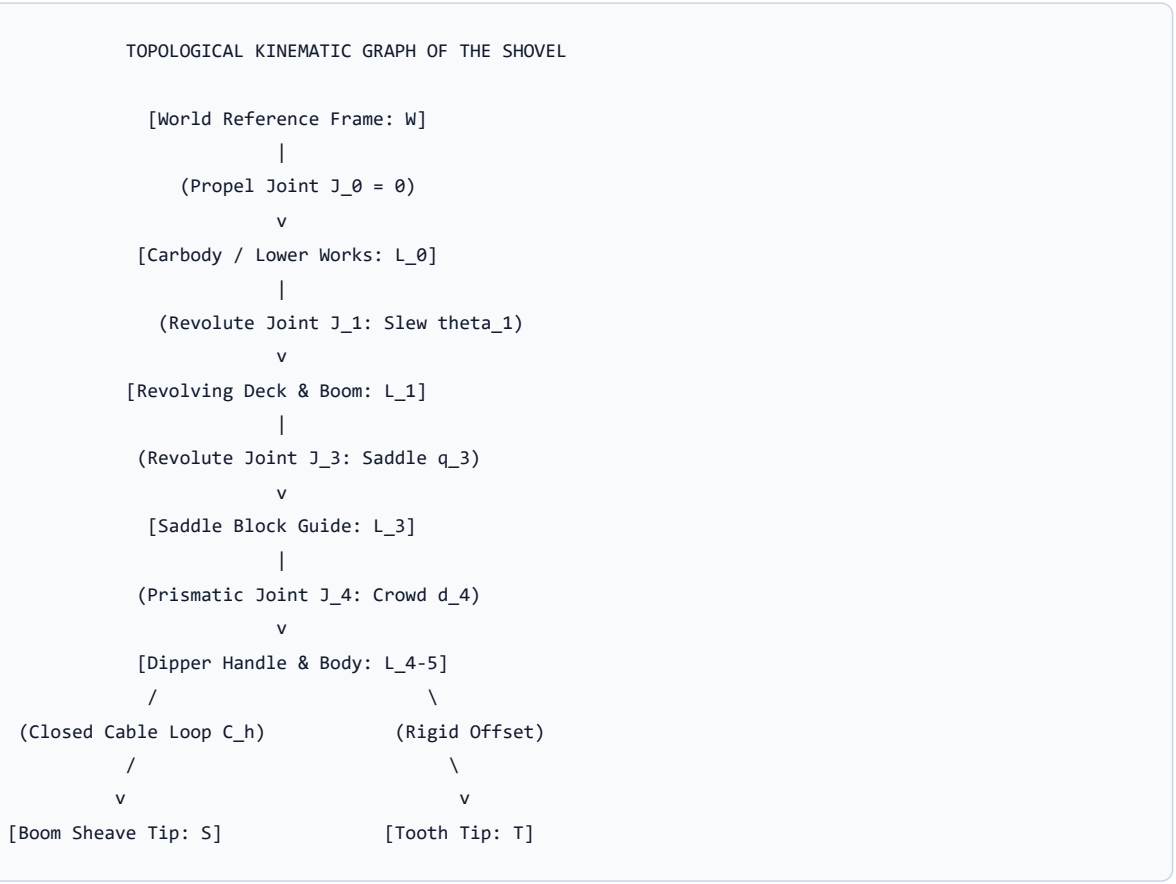
6.1.5 Subsystem 5: Dipper Assembly & Front-End Attachment

- **Dipper Bucket:** A high-strength fabricated alloy body with a cast manganese steel front lip designed to resist violent gouging and impact.
- **Ground Engaging Tools (GET):** Replaceable forged alloy-steel teeth and adapters keyed onto the lip.
- **Bail Pin & Equalizing Sheave:** The structural pin B connecting the hoist rope bail to the rear upper brackets of the dipper body.
- **Dipper Door & Trip Mechanism:** A hinged bottom door held closed by a spring-loaded mechanical latch bar. When dumping over a haul truck, an electric trip motor winds a cable that pulls the latch, allowing gravity to swing the door open.

6.2 Topological Graph Decomposition

To implement the kinematic architecture in a simulation code or digital twin harness, the machine linkage is represented as a **Directed Kinematic Tree Graph** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$:

- **Vertices $\mathcal{V} = \{L_0, L_1, L_2, L_3, L_4, L_5\}$:** Rigid body links.
- **Edges $\mathcal{E} = \{J_1, J_2, J_3, J_4, C_h\}$:** Kinematic joints and cable constraints.



6.2.1 Graph Connectivity Table

Joint ID	Predecessor Link	Successor Link	Joint Type	Generalized Coordinate	Physical Mechanism
J_1	Carbody (L_0)	Revolving Deck (L_1)	Revolute (R)	θ_1 (Swing)	Bull gear pinion mesh
J_2	Revolving Deck (L_1)	Boom (L_2)	Fixed (F)	Fixed angle $\alpha_b = 45^\circ$	Boom foot pins + bridge ropes
J_3	Boom (L_2)	Saddle Block (L_3)	Revolute (R)	q_3 (Elevation)	Shipper shaft bushings at O_3
J_4	Saddle Block (L_3)	Dipper Handle (L_4)	Prismatic (P)	d_4 (Crowd Extension)	Handle rack & shipper pinions
J_5	Handle (L_4)	Dipper Body (L_5)	Fixed / Pinned (F)	Fixed geometry (L_{tx}, L_{ty})	Pitch adjustment strut links
C_h	Boom Tip (S)	Dipper Bail (B)	Cable Constraint	Hoist Rope Length L_h	Dual wire ropes over sheaves

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6.3 Universal Reference Frame Tree & Coordinate System Definitions

We now construct a complete, standard set of right-handed orthogonal coordinate frames from the ground world frame to the dipper teeth.

```
                    REFERENCE FRAME TREE HIERARCHY

        {W} World Ground Frame
                 |
                 | (Position & Heading)
                 v
        {0} Revolving Deck Center
                 |
                 | [x_b, 0, y_b]^T
                 v
        {1} Boom Foot Pin O_1
                 |
                 | Rotated by alpha_b = 45 deg
                 v
        {2} Boom Axis Frame
                 |
                 | [L_b1, 0, 0]^T
                 v
        {3} Shipper Shaft Pivot O_3
                 |
                 | Rotated by saddle angle q_3
                 v
        {4} Saddle Block Frame
                 |
                 | Translated by crowd extension d_4
                 v
        {5} Dipper Handle End / Bail Pin B
                 |
                 | Fixed offset [L_tx, -L_ty, 0]^T
                 v
        {T} Dipper Tooth Tip Frame
```

6.3.1 Detailed Frame Specifications

1. **World Ground Frame $\{W\}$:**

- Origin O_W: Stationary benchmark point on the pit floor.
- Z_W: Vertical upward, opposing gravity ($\mathbf{g} = [0, 0, -g]^T$).
- X_W: Horizontal East.
- Y_W: Horizontal North.

1. **Revolving Deck Frame $\{0\}$:**

- Origin O_0: Intersection of the vertical machine center-of-rotation axis with the top plate of the revolving deck.
- Z_0: Vertical upward along the machine swing rotation axis ($Z_0 \parallel Z_W$).
- X_0: Horizontal forward along the machine centerline of symmetry.
- Y_0: Horizontal to the machine operator's left, completing the right-handed triad ($Y_0 = Z_0 \times X_0$).

1. **Boom Foot Pin Frame $\{1\}$:**

- Origin O_1: Center of the boom heel pivot pins $(x_{\text{boom}}, 0, y_{\text{boom}})_0$.
- X_1: Parallel to X_0 (horizontal forward).
- Y_1: Parallel to Y_0 (transverse left, along the boom pin axis).
- Z_1: Parallel to Z_0 (vertical upward).

1. **Boom Axis Frame $\{2\}$:**

- Origin O_1: Fixed at boom foot.
- Rotated about Y_1 by boom elevation angle α_b:
- X_2: Directed straight along the boom centerline toward the boom-point sheave center S.
- Y_2: Parallel to Y_1 (transverse left).
- Z_2: Perpendicular to boom axis, pointing upward-backward.

1. **Shipper Shaft / Saddle Pivot Frame $\{3\}$:**

- Origin O_3: Center of the shipper shaft at fixed distance L_{b1} along the boom axis X_2:

$$\mathbf{p}_{O_3/O_1} = \begin{matrix} L_{b1} \\ 0 \\ 0 \end{matrix}_2 \implies \mathbf{p}_{O_3/O_0} = \begin{matrix} x_{\text{boom}} + L_{b1}\cos\alpha_b \\ 0 \\ y_{\text{boom}} + L_{b1}\sin\alpha_b \end{matrix}_0 \tag{6.1}$$

- Y_3: Coincident with the shipper shaft axis of rotation.

1. **Saddle Block Frame $\{4\}$:**

- Origin O_3: Fixed at the shipper shaft center.
- Rotated about Y_3 by the absolute handle elevation angle q_3 with respect to horizontal X_0:
- X_4: Directed parallel to the dipper handle rack toward the dipper teeth.
- Y_4: Transverse along the shipper shaft axis.
- Z_4: Perpendicular to handle rack, pointing upward ($Z_4 = X_4 \times Y_4$).

1. **Dipper Bail Pin Frame $\{5\}$:**

- Origin B: Positioned at linear crowd extension d_4 along axis X_4:

$$\mathbf{p}_{B/O_3} = \begin{matrix} d_4 \\ 0 \\ 0 \end{matrix}_4 \tag{6.2}$$

1. **Tooth Tip Frame $\{T\}$:**

- Origin T: Located at the cutting edge of the central tooth tip.
- Rigidly fixed relative to Frame $\{5\}$ by longitudinal offset L_{tx} and downward offset L_{ty}:

$$\mathbf{p}_{T/B} = \begin{matrix} L_{tx} \\ 0 \\ -L_{ty} \end{matrix}_4 \implies \mathbf{p}_{T/O_3} = \begin{matrix} d_4 + L_{tx} \\ 0 \\ -L_{ty} \end{matrix}_4 \tag{6.3}$$

6.4 Comprehensive Machine Benchmark Parameter Audit: P&H 2100BL vs. TZ WK-20

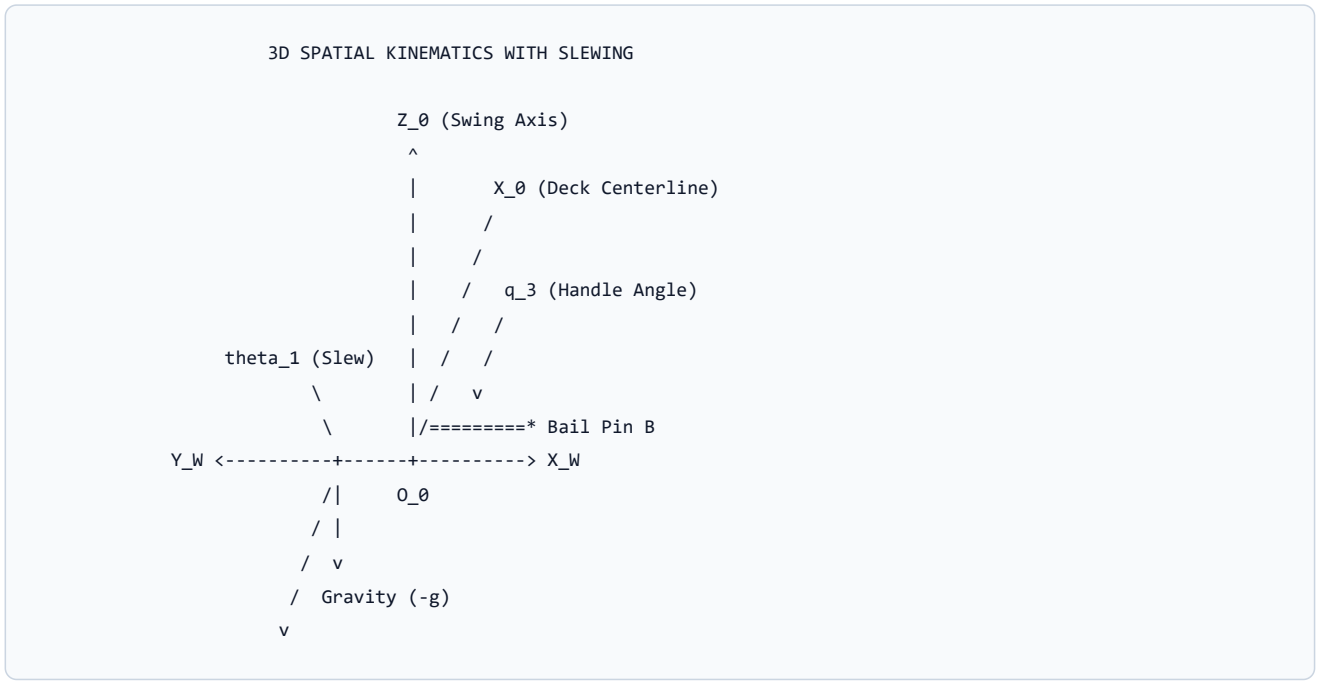
The following table provides the complete, authenticated parameter sets for the two machines investigated in our research:

Parameter Description	Mathematical Symbol	Engineering Units	Machine A: P&H 2100BL Benchmark	Machine B: TZ WK-20 Deployment	Primary Source Reference
Rated Bucket Payload	M_{payload}	t (kg)	$20.0\,\text{t}$ ($20,000\,\text{kg}$)	$55.0\,\text{t}$ ($55,000\,\text{kg}$)	[A] OEM Specification Sheets
Nominal Bucket Capacity	V_{bucket}	m^3	$13.8\,\text{m}^3$ ($18.0\,\text{yd}^3$)	$20.0\,\text{m}^3$ ($26.2\,\text{yd}^3$)	[A] Mine Operations Manual
Machine Total Working Weight	M_{total}	t	$345\,\text{t}$	$860\,\text{t}$	[A] Erection Manual
Boom Chord Total Length	L_b	m	$16.76\,\text{m}$ ($55\,\text{ft}$)	$21.50\,\text{m}$	[A] GA Assembly Drawing
Boom Fixed Elevation Angle	α_b	deg	45.0°	45.0°	[A] OEM Standard Setting
Boom Foot Coordinates	(x_b, y_b)	m	$(3.65, 2.10)\,\text{m}$	$(4.85, 2.80)\,\text{m}$	[A] Deck CAD Layout
Shipper Shaft Pivot Position	L_{b1}	m	$8.84\,\text{m}$	$11.20\,\text{m}$	[A] M1 Model Documentation
Boom-Point Sheave Radius	R_{sheave}	m	$0.914\,\text{m}$ ($72\,\text{in}$ dia)	$1.200\,\text{m}$ ($94.5\,\text{in}$ dia)	[A] Sheave Casting Drawing
Sheave Center Coordinates	(x_s, y_s)	m	$(15.50, 13.95)\,\text{m}$	$(20.05, 18.00)\,\text{m}$	[A] Kinematic Calculation
Dipper Handle Total Length	L_h	m	$11.89\,\text{m}$ ($39\,\text{ft}$)	$15.50\,\text{m}$	[A] Handle Structure Spec
Crowd Linear Stroke Limits	$[d_{4,\min}, d_{4,\max}]$	m	$[7.50, 14.50]\,\text{m}$	$[9.00, 18.20]\,\text{m}$	[A] Stop Block Physical Audit
Saddle Elevation	$[q_{3,\min}, q_{3,\max}]$	deg	$[-35.0^\circ, +48.0^\circ]$	$[-30.0^\circ, +50.0^\circ]$	[A] Joint Resolver

Parameter Description	Mathematical Symbol	Engineering Units	Machine A: P&H 2100BL Benchmark	Machine B: TZ WK-20 Deployment	Primary Source Reference
Angle Range					Envelope
Tooth Tip Long. Offset	L_{tx}	m	$2.00\,\mathrm{m}$	$2.50\,\mathrm{m}$	[A] Dipper Fabrication Print
Tooth Tip Down. Offset	L_{ty}	m	$1.00\,\mathrm{m}$	$1.20\,\mathrm{m}$	[A] Dipper Fabrication Print
Dipper Body Empty Mass	$m_{d,\mathrm{empty}}$	kg	$17,000\,\mathrm{kg}$	$44,000\,\mathrm{kg}$	[A] Certified Scale Weight
Dipper Handle Mass	m_h	kg	$11,500\,\mathrm{kg}$	$24,000\,\mathrm{kg}$	[A] Weight Certificate
Saddle Block Mass	m_s	kg	$3,500\,\mathrm{kg}$	$8,500\,\mathrm{kg}$	[A] Casting Data Sheet
Hoist Rope Diameter	d_{rope}	mm	$57.0\,\mathrm{mm}$ ($2.25\,\mathrm{in}$)	$72.0\,\mathrm{mm}$ ($2.83\,\mathrm{in}$)	[A] Wire Rope Spec Sheet
Hoist Reeving Multiplicity	n_{reeve}	$-$	2 (Twin lines, direct)	2 (Twin lines, direct)	[A] Reeving Diagram
Maximum Hoist Line Pull	$F_{h,\max}$	kN	$750\,\mathrm{kN}$	$1,650\,\mathrm{kN}$	[A] Motor Torque / Gear Limit
Maximum Crowd Push Force	$F_{c,\max}$	kN	$350\,\mathrm{kN}$	$800\,\mathrm{kN}$	[A] Pinion Tooth Bending Limit
Nominal Swing Slew Speed	$\dot{\theta}_{1,\max}$	rpm	$3.50\,\mathrm{rpm}$ ($0.366\,\mathrm{rad/s}$)	$2.80\,\mathrm{rpm}$ ($0.293\,\mathrm{rad/s}$)	[A] Motor VFD Limit

6.5 Full 3D Spatial Forward Kinematics with Slewing

We now formulate the complete 3D homogeneous transformation matrices from the world frame $\{W\}$ to the dipper tooth tip $\{T\}$.



6.5.1 Transformation from World $\{W\}$ to Revolving Frame $\{0\}$

Let the shovel be positioned at ground Cartesian coordinates (X_c, Y_c, Z_c) with upper deck slewing angle $\theta_1(t)$ measured relative to the world X_W axis:

$$\mathbf{T}_0^W = \begin{matrix} \cos\theta_1 & -\sin\theta_1 & 0 & X_c \\ \sin\theta_1 & \cos\theta_1 & 0 & Y_c \\ 0 & 0 & 1 & Z_c \\ 0 & 0 & 0 & 1 \end{matrix} \tag{6.4}$$

6.5.2 Position Vector of the Shipper Shaft Pivot O_3 in Frame $\{0\}$

From Equation (6.1), because the boom foot is fixed to the deck in the symmetry plane ($Y_0 = 0$):

$$\mathbf{p}_{O_3}^0 = \begin{matrix} x_3 \\ 0 \\ z_3 \end{matrix} = \begin{matrix} x_b + L_{b1}\cos\alpha_b \\ 0 \\ z_b + L_{b1}\sin\alpha_b \end{matrix} \tag{6.5}$$

6.5.3 Transformation from Frame $\{0\}$ to Saddle Block Frame $\{4\}$

The saddle block frame $\{4\}$ is translated to O_3 and pitched by handle angle q_3 about the transverse Y_0 axis:

$$\mathbf{T}_4^0 = \begin{matrix} \cos q_3 & 0 & \sin q_3 & x_3 \\ 0 & 1 & 0 & 0 \\ -\sin q_3 & 0 & \cos q_3 & z_3 \\ 0 & 0 & 0 & 1 \end{matrix} \tag{6.6}$$

(Notice: In standard robotic coordinate conventions, pitching upward about Y transforms X into Z. In standard planar mechanics where q_3 is defined from horizontal X toward vertical Z, $\mathbf{p} = [r\cos q_3, 0, r\sin q_3]^T$).

6.5.4 3D Cartesian Position of Dipper Tooth Tip T

In the handle frame $\{4\}$, the tooth tip is located at:

$$\mathbf{p}_T^4 = \begin{matrix} d_4 + L_{tx} \\ 0 \\ -L_{ty} \end{matrix} \tag{6.7}$$

Transforming into the revolving deck frame $\{0\}$:

$$\mathbf{p}_T^0 = \begin{matrix} x_T^0 \\ y_T^0 \\ z_T^0 \end{matrix} = \begin{matrix} x_3 + (d_4 + L_{tx})\cos q_3 + L_{ty}\sin q_3 \\ 0 \\ z_3 + (d_4 + L_{tx})\sin q_3 - L_{ty}\cos q_3 \end{matrix} \tag{6.8}$$

Transforming into the global world ground frame $\{W\}$ via Equation (6.4):

$$\begin{matrix} x_T^W \\ y_T^W \\ z_T^W \end{matrix} = \begin{matrix} X_c \\ Y_c \\ Z_c \end{matrix} + \begin{matrix} \cos\theta_1 & -\sin\theta_1 & 0 & x_T^0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & z_T^0 \end{matrix} \tag{6.9}$$

Carrying out the matrix multiplication yields the **Universal 3D Forward Kinematics of the Tooth Tip**:

$$x_T^W(\theta_1, q_3, d_4) = X_c + x_T^0(q_3, d_4)\cos\theta_1 \tag{6.10a}$$

$$y_T^W(\theta_1, q_3, d_4) = Y_c + x_T^0(q_3, d_4)\sin\theta_1 \tag{6.10b}$$

$$z_T^W(q_3, d_4) = Z_c + z_T^0(q_3, d_4) \tag{6.10c}$$

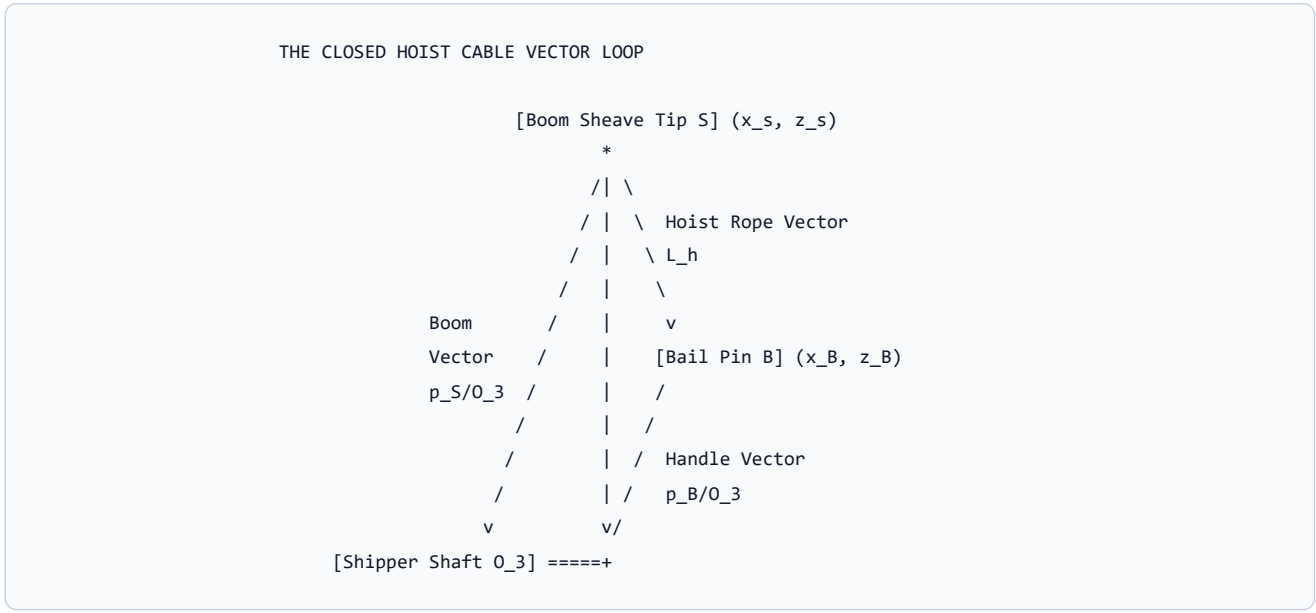
IMPORTANT

PHYSICAL INSIGHT INTO SPATIAL SHOVEL MOTIONS

1. Slewing angle θ_1 rotates the entire vertical digging plane around the machine kingpin Z_0.
2. The vertical elevation z_T^W is **completely independent of the swing angle θ_1**. It depends solely on crowd extension d_4 and saddle elevation q_3.
3. The horizontal radius from the swing axis to the dipper teeth is precisely $R_{\text{slew}} = x_T^0(q_3, d_4)$.

6.6 The Closed Hoist Cable Loop: Geometry, Length & Moment Arm

A defining structural characteristic of an electric rope shovel is that the dipper is hoisted by flexible wire cables passing over boom-point sheaves. This forms a closed kinematic loop between the boom, dipper, and saddle block.



6.6.1 Vector Loop Formulation

Let $S = (x_s, z_s)$ denote the fixed coordinates of the boom-point sheave center in the revolving frame $\{0\}$:

$$x_s = x_b + L_b \cos \alpha_b \tag{6.11a}$$

$$z_s = z_b + L_b \sin \alpha_b \tag{6.11b}$$

Let $B = (x_B, z_B)$ denote the coordinates of the dipper bail pin in Frame $\{0\}$:

$$x_B(q_3, d_4) = x_3 + d_4 \cos q_3 \tag{6.12a}$$

$$z_B(q_3, d_4) = z_3 + d_4 \sin q_3 \tag{6.12b}$$

Define the **Hoist Cable Vector** $\mathbf{L}_h$ extending from the boom sheave center S to the bail pin B:

$$\mathbf{L}_h = \mathbf{p}_B^0 - \mathbf{p}_S^0 = \begin{bmatrix} x_B - x_s \\ z_B - z_s \end{bmatrix} = \begin{bmatrix} (x_3 - x_s) + d_4 \cos q_3 \\ (z_3 - z_s) + d_4 \sin q_3 \end{bmatrix} \tag{6.13}$$

6.6.2 Analytical Closed-Form Hoist Rope Length $L_h(q_3, d_4)$

The instantaneous straight-line suspended length of the hoist rope is the Euclidean norm of $\mathbf{L}_h$:

$$L_h(q_3, d_4) = \|\mathbf{L}_h\| = \sqrt{(x_B - x_s)^2 + (z_B - z_s)^2} \tag{6.14}$$

Substitute the vector components from Equation (6.13):

$$L_h^2 = \left[(x_3 - x_s) + d_4 \cos q_3\right]^2 + \left[(z_3 - z_s) + d_4 \sin q_3\right]^2 \tag{6.15}$$

Expanding the squared binomials:

$$L_h^2 = (x_3 - x_s)^2 + 2d_4(x_3 - x_s)\cos q_3 + d_4^2 \cos^2 q_3 + (z_3 - z_s)^2 + 2d_4(z_3 - z_s)\sin q_3 +$$

Notice that $(x_3 - x_s)^2 + (z_3 - z_s)^2 = (L_b - L_{b1})^2$ is the square of the distance along the boom from saddle pivot O_3 to sheave center S. Let:

$$L_{b2} = L_b - L_{b1} \tag{6.16}$$

Furthermore, $d_4^2(\cos^2 q_3 + \sin^2 q_3) = d_4^2$. Therefore:

$$L_h(q_3, d_4) = \sqrt{L_{b2}^2 + d_4^2 + 2d_4\left[(x_3 - x_s)\cos q_3 + (z_3 - z_s)\sin q_3\right]} \tag{6.17}$$

Equation (6.17) is the **exact, closed-form algebraic formula for suspended hoist rope length**.

6.6.3 Geometric Hoist Moment Arm $h_h(q_3, d_4)$

The tension force F_{hoist} in the wire ropes creates a lifting torque τ_{hoist} about the shipper shaft pivot O_3:

$$\tau_{\text{hoist}} = F_{\text{hoist}} \cdot h_h(q_3, d_4) \tag{6.18}$$

where $h_h(q_3, d_4)$ is the perpendicular moment arm from pivot O_3 to the line of action of the hoist cable.

By vector mechanics, the torque about O_3 produced by tension force $\mathbf{F}_h = F_{\text{hoist}} \frac{\mathbf{p}_S^0 - \mathbf{p}_B^0}{\|\mathbf{p}_S^0 - \mathbf{p}_B^0\|} = -F_{\text{hoist}} \frac{\mathbf{L}_h}{L_h}$ is:

$$\boldsymbol{\tau}_{\text{hoist}} = \mathbf{p}_{B/O_3} \times \left(-F_{\text{hoist}} \frac{\mathbf{L}_h}{L_h}\right) \tag{6.19}$$

The scalar moment arm h_h is the magnitude of the 2D cross-product:

$$h_h(q_3, d_4) = \frac{1}{L_h}\,\mathbf{p}_{B/O_3} \times (-\mathbf{L}_h) \;= \frac{1}{L_h}\,\mathbf{p}_{B/O_3} \times (\mathbf{p}_S^0 - \mathbf{p}_B^0) \tag{6.20}$$

Since $\mathbf{p}_{B/O_3} \times \mathbf{p}_B^0 = \mathbf{p}_{B/O_3} \times (\mathbf{p}_{O_3} + \mathbf{p}_{B/O_3}) = \mathbf{p}_{B/O_3} \times \mathbf{p}_{O_3}$:

$$h_h(q_3, d_4) = \frac{1}{L_h}\,x_{B/O_3}(z_s - z_3) - z_{B/O_3}(x_s - x_3) \tag{6.21}$$

Substitute $x_{B/O_3} = d_4 \cos q_3$ and $z_{B/O_3} = d_4 \sin q_3$:

$$h_h(q_3, d_4) = \frac{d_4}{L_h(q_3, d_4)} \left| (z_s - z_3)\cos q_3 - (x_s - x_3)\sin q_3 \right| \tag{6.22}$$

> **TIP**
>
> ### PHYSICAL INTUITION ON HOIST MOMENT ARM
>
> 1. Because $(x_s - x_3) = L_{b2}\cos\alpha_b$ and $(z_s - z_3) = L_{b2}\sin\alpha_b$:
>
> $$(z_s - z_3)\cos q_3 - (x_s - x_3)\sin q_3 = L_{b2}\left(\sin\alpha_b \cos q_3 - \cos\alpha_b \sin q_3\right) = L_{b2}\sin(\alpha_b - q_3)$$
>
> 2. Substituting into Equation (6.22) yields the remarkably elegant formula:
>
> $$h_h(q_3, d_4) = \frac{d_4 L_{b2}\sin(\alpha_b - q_3)}{L_h(q_3, d_4)}$$
>
> 3. **Maximum Lifting Advantage:** The moment arm peaks when the dipper handle is perpendicular to the boom ($\alpha_b - q_3 \approx 90^\circ$). When the dipper is hoisted near the boom tip ($\alpha_b - q_3 \to 0^\circ$), the moment arm collapses toward zero!

6.7 Five Fully Solved Research Practice Problems

Problem 6.1: Complete 3D Tooth Tip Forward Kinematics (P&H 2100BL)

Problem Statement:

A P&H 2100BL shovel (Machine A) is situated at pit bench coordinates $(X_c, Y_c, Z_c) = (120.0, 450.0, 85.0)\,\text{m}$.

The joint resolvers read:

- Upper deck slew angle: $\theta_1 = 45.0^\circ$ ($\frac{\pi}{4}$ rad)
- Saddle block angle: $q_3 = 15.0^\circ$ ($0.2618\,\text{rad}$)
- Crowd handle extension: $d_4 = 11.50\,\text{m}$

Using the P&H 2100BL parameters from Section 6.4 ($x_b = 3.65\,\text{m}, z_b = 2.10\,\text{m}, L_{b1} = 8.84\,\text{m}, \alpha_b = 45^\circ, L_{tx} = 2.00\,\text{m}, L_{ty} = 1.00\,\text{m}$):

1. Compute the Cartesian coordinates (x_3, z_3) of the shipper shaft pivot O_3 in Frame $\{0\}$.
2. Compute the dipper tooth tip position (x_T^0, z_T^0) in the revolving frame $\{0\}$.
3. Compute the global 3D world coordinates (x_T^W, y_T^W, z_T^W) of the tooth tip.

Step-by-Step Solution:

Step 1: Compute shipper shaft pivot O_3 in Frame $\{0\}$

$$\cos 45^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2} \approx 0.707107$$

$$x_3 = x_b + L_{b1}\cos 45^\circ = 3.65 + (8.84)(0.707107) = 3.65 + 6.2508 = 9.9008\,\text{m}$$

$$z_3 = z_b + L_{b1}\sin 45^\circ = 2.10 + (8.84)(0.707107) = 2.10 + 6.2508 = 8.3508\,\text{m}$$

$$\mathbf{p}_{O_3}^0 = \begin{matrix} 9.9008 \\ 0 \\ 8.3508 \end{matrix}\ \text{m}$$

Step 2: Compute tooth tip position in Frame $\{0\}$

$$r_T = d_4 + L_{tx} = 11.50 + 2.00 = 13.50\,\text{m}$$

$$\cos 15^\circ \approx 0.965926, \quad \sin 15^\circ \approx 0.258819$$

Using Equation (6.8):

$$x_T^0 = x_3 + r_T \cos q_3 + L_{ty} \sin q_3$$

$$x_T^0 = 9.9008 + (13.50)(0.965926) + (1.00)(0.258819)$$

$$x_T^0 = 9.9008 + 13.0400 + 0.2588 = 23.1996\,\mathrm{m}$$

$$z_T^0 = z_3 + r_T \sin q_3 - L_{ty} \cos q_3$$

$$z_T^0 = 8.3508 + (13.50)(0.258819) - (1.00)(0.965926)$$

$$z_T^0 = 8.3508 + 3.4941 - 0.9659 = 10.8790\,\mathrm{m}$$

$$\mathbf{p}_T^0 = \begin{matrix} 23.200 \\ 0 \\ 10.879 \end{matrix} \ \mathrm{m}$$

(The dipper tooth tip reaches a horizontal radius of $23.20\,\mathrm{m}$ from the machine swing center and an elevation of $10.88\,\mathrm{m}$ above the deck).

Step 3: Transform into World Ground Frame $\{W\}$

Using Equation (6.10):

$$\cos\theta_1 = \cos 45^\circ \approx 0.707107, \quad \sin\theta_1 = \sin 45^\circ \approx 0.707107$$

$$x_T^W = X_c + x_T^0 \cos 45^\circ = 120.0 + (23.1996)(0.707107) = 120.0 + 16.4046 = 136.405\,\text{m}$$

$$y_T^W = Y_c + x_T^0 \sin 45^\circ = 450.0 + (23.1996)(0.707107) = 450.0 + 16.4046 = 466.405\,\text{m}$$

$$z_T^W = Z_c + z_T^0 = 85.0 + 10.8790 = 95.879\,\text{m}$$

$$\mathbf{p}_T^W = \begin{matrix} 136.405\,\text{m} \\ 466.405\,\text{m} \\ 95.879\,\text{m} \end{matrix}$$

Problem 6.2: Hoist Cable Length and Moment Arm on TZ WK-20 Heavy Shovel

Problem Statement:

On a TZ WK-20 excavator (Machine B), the parameters are:

- Boom chord length: $L_b = 21.50\,\text{m}$, inclination $\alpha_b = 45.0^\circ$
- Boom foot coordinates: $(x_b, z_b) = (4.85, 2.80)\,\text{m}$
- Shipper shaft pivot distance: $L_{b1} = 11.20\,\text{m}$
- Current joint state: $q_3 = 20.0^\circ$, $d_4 = 13.00\,\text{m}$

1. Compute the boom-point sheave center coordinates (x_s, z_s) and shipper shaft pivot (x_3, z_3).
2. Calculate the suspended length of the hoist rope $L_h(q_3, d_4)$.

3. Calculate the perpendicular hoist moment arm $h_h(q_3, d_4)$ about the shipper shaft pivot O_3.
4. If the twin hoist ropes sustain a combined tension of $F_{\text{hoist}} = 1,200\,\text{kN}$, compute the hoist lifting torque τ_{hoist} applied to the saddle block.

Step-by-Step Solution:

Step 1: Compute sheave center S and pivot O_3

$$\cos 45^\circ = \sin 45^\circ \approx 0.707107$$

$$x_s = x_b + L_b \cos 45^\circ = 4.85 + (21.50)(0.707107) = 4.85 + 15.2028 = 20.0528\,\text{m}$$

$$z_s = z_b + L_b \sin 45^\circ = 2.80 + (21.50)(0.707107) = 2.80 + 15.2028 = 18.0028\,\text{m}$$

$$x_3 = x_b + L_{b1} \cos 45^\circ = 4.85 + (11.20)(0.707107) = 4.85 + 7.9196 = 12.7696\,\text{m}$$

$$z_3 = z_b + L_{b1} \sin 45^\circ = 2.80 + (11.20)(0.707107) = 2.80 + 7.9196 = 10.7196\,\text{m}$$

$$L_{b2} = L_b - L_{b1} = 21.50 - 11.20 = 10.30\,\text{m}$$

Step 2: Compute suspended hoist rope length L_h

Coordinates of bail pin B at $q_3 = 20^\circ, d_4 = 13.0\,\text{m}$:

$$\cos 20^\circ \approx 0.939693, \quad \sin 20^\circ \approx 0.342020$$

$$x_B = x_3 + d_4 \cos 20^\circ = 12.7696 + (13.0)(0.939693) = 12.7696 + 12.2160 = 24.9856\,\text{m}$$

$$z_B = z_3 + d_4 \sin 20^\circ = 10.7196 + (13.0)(0.342020) = 10.7196 + 4.4463 = 15.1659\,\text{m}$$

Vector from sheave S to bail B:

$$\Delta x = x_B - x_s = 24.9856 - 20.0528 = +4.9328\,\text{m}$$

$$\Delta z = z_B - z_s = 15.1659 - 18.0028 = -2.8369\,\text{m}$$

$$L_h = \sqrt{(\Delta x)^2 + (\Delta z)^2} = \sqrt{(4.9328)^2 + (-2.8369)^2} = \sqrt{24.3325 + 8.0480} = \sqrt{32.380}$$

Result: The suspended cable length between boom sheave and bail pin is **$L_h = 5.690\,\text{m}$**.

Step 3: Compute hoist moment arm h_h

Using the formula derived in Section 6.6.3:

$$h_h(q_3, d_4) = \frac{d_4 L_{b2} \sin(\alpha_b - q_3)}{L_h}$$

Here:

$$\alpha_b - q_3 = 45.0^\circ - 20.0^\circ = 25.0^\circ \implies \sin 25.0^\circ \approx 0.422618$$

$$h_h = \frac{(13.00\,\text{m})(10.30\,\text{m})(0.422618)}{5.6904\,\text{m}} = \frac{56.5886}{5.6904} \approx 9.9446\,\text{m}$$

Result: The hoist line pull moment arm about the shipper shaft is **$h_h = 9.945\,\text{m}$**.

Step 4: Compute hoist lifting torque

$$\tau_{\text{hoist}} = F_{\text{hoist}} \cdot h_h = (1,200\,\text{kN})(9.9446\,\text{m}) = 11,933.5\,\text{kN}\cdot\text{m} \approx +11.934\,\text{MN}\cdot\text{m}$$

Result: The hoist cables deliver **$+11.934\,\text{MN}\cdot\text{m}$** of lifting torque to elevate the $119\,\text{t}$ front-end assembly.

Problem 6.3: Spatial 3D Velocity Vector under Combined Slew-Crowd-Hoist Motion

Problem Statement:

On a P&H 2100BL shovel, all three primary motions are engaged simultaneously during a high-speed bank exit:

- Slew speed: $\dot{\theta}_1 = 0.25\,\text{rad/s}$ ($2.39\,\text{rpm}$)
- Saddle pitch speed: $\dot{q}_3 = 0.15\,\text{rad/s}$
- Crowd extension speed: $\dot{d}_4 = 0.80\,\text{m/s}$

Current posture: $\theta_1 = 30^\circ$, $q_3 = 20^\circ$, $d_4 = 11.0\,\text{m}$, with tooth offsets $r_T = 13.0\,\text{m}, L_{ty} = 1.0\,\text{m}, x_3 = 9.90\,\text{m}, z_3 = 8.35\,\text{m}$.

1. Compute the planar velocities $(\dot{x}_T^0, \dot{z}_T^0)$ in the deck frame.
2. Formulate the full 3D Cartesian velocity vector $\mathbf{v}_T^W = [\dot{x}_T^W, \dot{y}_T^W, \dot{z}_T^W]^T$ in the ground frame.
3. Determine the tangential slewing velocity component and the resultant spatial 3D speed $\|\mathbf{v}_T^W\|$.

Step-by-Step Solution:

Step 1: Compute planar velocities in Frame $\{0\}$

Using the tooth tip Jacobian from Chapter 2 (Equation (2.33)):

$$J_{11} = -r_T \sin q_3 + L_{ty} \cos q_3 = -(13.0)(0.342020) + (1.0)(0.939693) = -4.4463 + 0.9$$

$$J_{12} = \cos q_3 = 0.939693$$

$$J_{21} = r_T \cos q_3 + L_{ty} \sin q_3 = (13.0)(0.939693) + (1.0)(0.342020) = 12.2160 + 0.3420$$

$$J_{22} = \sin q_3 = 0.342020$$

$$\dot{x}_T^0 = J_{11}\dot{q}_3 + J_{12}\dot{d}_4 = (-3.5066)(0.15) + (0.939693)(0.80) = -0.5260 + 0.7518 = +0$$

$$\dot{z}_T^0 = J_{21}\dot{q}_3 + J_{22}\dot{d}_4 = (12.5580)(0.15) + (0.342020)(0.80) = 1.8837 + 0.2736 = +2.15$$

Also compute horizontal radius x_T^0:

$$x_T^0 = x_3 + r_T \cos q_3 + L_{ty} \sin q_3 = 9.90 + (13.0)(0.939693) + (1.0)(0.342020) = 9.90 +$$

Step 2: Compute full 3D velocity in World Ground Frame $\{W\}$

Differentiating Equation (6.10) with respect to time:

$$\dot{x}_T^W = \dot{x}_T^0 \cos\theta_1 - x_T^0 \dot{\theta}_1 \sin\theta_1$$

$$\dot{y}_T^W = \dot{x}_T^0 \sin\theta_1 + x_T^0 \dot{\theta}_1 \cos\theta_1$$

$$\dot{z}_T^W = \dot{z}_T^0$$

Evaluate at $\theta_1 = 30^\circ$ ($\cos 30^\circ \approx 0.866025, \sin 30^\circ = 0.500000$):

$$\text{Tangential slewing speed } v_{\text{slew}} = x_T^0 \dot{\theta}_1 = (22.4580\,\text{m})(0.25\,\text{rad/s}) = 5.6145\,\text{m/s}$$

$$\dot{x}_T^W = (0.2258)(0.866025) - (5.6145)(0.500000) = 0.1955 - 2.8073 = -2.6118\,\text{m/s}$$

$$\dot{y}_T^W = (0.2258)(0.500000) + (5.6145)(0.866025) = 0.1129 + 4.8623 = +4.9752\,\text{m/s}$$

$$\dot{z}_T^W = +2.1573\,\text{m/s}$$

$$\mathbf{v}_T^W = \begin{matrix} -2.612 \\ +4.975 \\ +2.157 \end{matrix} \quad \text{m/s}$$

Step 3: Total 3D Speed

$$\|\mathbf{v}_T^W\| = \sqrt{(-2.6118)^2 + (4.9752)^2 + (2.1573)^2} = \sqrt{6.8215 + 24.7526 + 4.6539} = \sqrt{36}$$

(Engineering Insight: Notice that the slewing motion contributes the vast majority of the spatial kinetic energy: $v_{\text{slew}} = 5.61\,\text{m/s}$ out of the total $6.02\,\text{m/s}$ speed! This demonstrates why slewing collisions with highwalls or haul trucks are so destructive).

Problem 6.4: Reeving Kinematics & Hoist Drum Angular Velocity

Problem Statement:

On a TZ WK-20 excavator, the hoist drum has an outer pitch diameter of $D_{\text{drum}} = 1.60\,\text{m}$ (radius $R_{\text{drum}} = 0.80\,\text{m}$).

The hoist motor drives the drum through a double-reduction helical gearbox with overall gear ratio $i_{\text{hoist}} = 32.5 : 1$.

The twin hoist cables are reeved in a 2-part direct configuration ($n_{\text{reeve}} = 2$).

At an instant during high-speed hoisting, the geometric cable length changes at rate $\dot{L}_h = -1.25\,\text{m/s}$ (winding in).

1. Calculate the linear line speed v_{cable} at which the wire rope winds onto the hoist drum.
2. Compute the rotational speed of the hoist drum ω_{drum} in rad/s and rpm.
3. Compute the required angular speed ω_{motor} of the dual $1,200\,\text{kW}$ AC hoist motors in rpm.

Step-by-Step Solution:

Step 1: Compute rope winding speed

For a 2-part reeving system where two parallel ropes hoist the bail directly:

$$v_{\text{cable}} = |\dot{L}_h| = 1.25\,\text{m/s}$$

Step 2: Compute hoist drum rotational speed

$$v_{\text{cable}} = R_{\text{drum}} \cdot \omega_{\text{drum}} \implies \omega_{\text{drum}} = \frac{v_{\text{cable}}}{R_{\text{drum}}} = \frac{1.25\,\text{m/s}}{0.80\,\text{m}} = 1.5625\,\text{rad/s}$$

Convert to revolutions per minute (rpm):

$$N_{\text{drum}} = \omega_{\text{drum}} \times \frac{60}{2\pi} = 1.5625 \times 9.5493 \approx 14.92\,\text{rpm}$$

Step 3: Compute electric motor shaft speed

$$N_{\text{motor}} = i_{\text{hoist}} \cdot N_{\text{drum}} = (32.5)(14.9208\,\text{rpm}) \approx 484.93\,\text{rpm}$$

$$\omega_{\text{motor}} = (1.5625\,\text{rad/s})(32.5) \approx 50.78\,\text{rad/s}$$

Result: The hoist motors spin at **$484.9\,\text{rpm}$** ($50.78\,\text{rad/s}$), well within their $0 - 1{,}000\,\text{rpm}$ constant-torque operating envelope.

Problem 6.5: Dipper Door Latch Disengagement Kinematics

Problem Statement:

When dumping payload into a $240\,\text{t}$ haul truck, the dipper bucket is held stationary at $q_3 = 35.0^\circ$, $d_4 = 12.0\,\text{m}$.

The dipper door has mass $m_{\text{door}} = 4{,}200\,\text{kg}$ and rotates about hinge pins located at the bottom rear of the bucket body. The distance from the door hinge to its center of mass is $r_{\text{door}} = 1.40\,\text{m}$. In the latched position, the door plate is inclined at angle $\phi_{\text{door}} = 20.0^\circ$ below the handle axis.

1. Determine the absolute orientation of the dipper door with respect to horizontal ground.
2. Compute the gravitational unlatching torque $\tau_{\text{gravity}}^{\text{door}}$ tending to swing the door open the instant the latch bar is retracted.

3. If an electric trip motor retracts the latch pin in $80\,\text{ms}$, compute the initial angular acceleration $\ddot{\phi}_{\text{door}}$ of the opening door (door inertia about hinge: $I_{\text{door}} = 6,500\,\text{kg}\cdot\text{m}^2$).

Step-by-Step Solution:

Step 1: Compute absolute door orientation

The handle is elevated at $q_3 = +35.0^\circ$. The door is offset downward by $\phi_{\text{door}} = 20.0^\circ$.

The absolute elevation angle of the door plate relative to horizontal is:

$$\theta_{\text{door}} = q_3 - \phi_{\text{door}} = 35.0^\circ - 20.0^\circ = +15.0^\circ$$

Step 2: Compute gravitational unlatching torque

The gravitational torque acting to swing the door open about its hinge is:

$$\tau_{\text{gravity}}^{\text{door}} = m_{\text{door}} g r_{\text{door}} \cos\theta_{\text{door}}$$

$$\tau_{\text{gravity}}^{\text{door}} = (4,200\,\text{kg})(9.81\,\text{m/s}^2)(1.40\,\text{m})\cos 15.0^\circ$$

$$\tau_{\text{gravity}}^{\text{door}} = (57,682.8\,\text{N}\cdot\text{m})(0.965926) \approx 55,717.3\,\text{N}\cdot\text{m} \approx +55.72\,\text{kN}\cdot\text{m}$$

Result: A massive gravitational torque of **$55.72\,\text{kN}\cdot\text{m}$** flings the door open the instant the latch releases.

Step 3: Initial angular acceleration

$$\ddot{\phi}_{\text{door}} = \frac{\tau_{\text{gravity}}^{\text{door}}}{I_{\text{door}}} = \frac{55,717.3\,\text{N}\cdot\text{m}}{6,500\,\text{kg}\cdot\text{m}^2} \approx 8.5719\,\text{rad/s}^2$$

Convert to degrees per second squared:

$$\ddot{\phi}_{\text{door}} = (8.5719) \times \frac{180}{\pi} \approx 491.1^{\circ}/\text{s}^2$$

*(Result: The door accelerates open at **$491^{\circ}/\text{s}^2$**, discharging $55\ \text{t}$ of rock into the truck tray in under $1.2\ \text{seconds}$!).*

6.8 MATLAB / Python Topological Simulator Script

```
% ========================================================================
% CHAPTER 6: 3D SPATIAL KINEMATICS & CABLE LOOP SIMULATOR
% Compatible with MATLAB R2025b / Octave 7+
% ========================================================================
function shovel_spatial_kinematics_demo()
    clc; clear; close all;

    % Select Machine: 'PH2100BL' or 'TZ_WK20'
    machine_type = 'TZ_WK20';
    p = load_machine_parameters(machine_type);

    fprintf('=== SIMULATING %s SPATIAL KINEMATICS ===\n', machine_type);

    % Operating Joint States
    theta1 = deg2rad(35.0); % Swing angle [rad]
    q3     = deg2rad(20.0); % Saddle angle [rad]
    d4     = 13.00;         % Crowd extension [m]

    % Velocities
    theta1_d = 0.20;        % Swing speed [rad/s]
    q3_d     = 0.15;        % Saddle speed [rad/s]
    d4_d     = 0.85;        % Crowd speed [m/s]

    % 1. Forward Kinematics Computation
    [p_T_world, p_B_deck, p_O3_deck, p_S_deck] = compute_3d_fk(theta1, q3, d4, p);

    fprintf('Shipper Shaft Pivot O3:  [X, Z] = [%.3f, %.3f] m\n', p_O3_deck(1), p_O3_deck(3));
    fprintf('Boom Point Sheave S:     [X, Z] = [%.3f, %.3f] m\n', p_S_deck(1), p_S_deck(3));
    fprintf('Dipper Bail Pin B:       [X, Z] = [%.3f, %.3f] m\n', p_B_deck(1), p_B_deck(3));
    fprintf('Global 3D Tooth Tip T:   [X, Y, Z] = [%.3f, %.3f, %.3f] m\n\n', ...
            p_T_world(1), p_T_world(2), p_T_world(3));

    % 2. Hoist Cable Loop Analysis
    [L_h, h_h] = compute_hoist_loop(q3, d4, p);
    fprintf('=== HOIST CABLE LOOP CHARACTERISTICS ===\n');
    fprintf('Suspended Cable Length L_h: %.4f m\n', L_h);
    fprintf('Hoist Moment Arm h_h:       %.4f m\n\n', h_h);

    % 3. 3D Velocity Vector
    v_T_world = compute_3d_velocity(theta1, q3, d4, theta1_d, q3_d, d4_d, p);
    fprintf('=== 3D CARTESIAN VELOCITY VECTOR ===\n');
    fprintf('v_x = %+.4f m/s\n', v_T_world(1));
    fprintf('v_y = %+.4f m/s\n', v_T_world(2));
    fprintf('v_z = %+.4f m/s\n', v_T_world(3));
    fprintf('Resultant Spatial Speed ||v|| = %.4f m/s\n', norm(v_T_world));
end

function p = load_machine_parameters(type)
    if strcmp(type, 'PH2100BL')
        p.xb = 3.65; p.zb = 2.10; p.Lb = 16.76; p.Lb1 = 8.84;
        p.alpha_b = deg2rad(45.0); p.Ltx = 2.00; p.Lty = 1.00;
    else % TZ WK-20
        p.xb = 4.85; p.zb = 2.80; p.Lb = 21.50; p.Lb1 = 11.20;
```

```
        p.alpha_b = deg2rad(45.0); p.Ltx = 2.50; p.Lty = 1.20;
    end
end

function [p_T_w, p_B_0, p_O3_0, p_S_0] = compute_3d_fk(th1, q3, d4, p)
    % Shipper Shaft Pivot O3
    x3 = p.xb + p.Lb1 * cos(p.alpha_b);
    z3 = p.zb + p.Lb1 * sin(p.alpha_b);
    p_O3_0 = [x3; 0; z3];

    % Boom Sheave S
    xs = p.xb + p.Lb * cos(p.alpha_b);
    zs = p.zb + p.Lb * sin(p.alpha_b);
    p_S_0 = [xs; 0; zs];

    % Bail Pin B
    xB = x3 + d4 * cos(q3);
    zB = z3 + d4 * sin(q3);
    p_B_0 = [xB; 0; zB];

    % Tooth Tip in Frame {0}
    rT = d4 + p.Ltx;
    xT0 = x3 + rT * cos(q3) + p.Lty * sin(q3);
    zT0 = z3 + rT * sin(q3) - p.Lty * cos(q3);

    % Global 3D (assuming ground frame coincident at origin)
    p_T_w = [xT0 * cos(th1);
             xT0 * sin(th1);
             zT0];
end

function [L_h, h_h] = compute_hoist_loop(q3, d4, p)
    x3 = p.xb + p.Lb1 * cos(p.alpha_b);
    z3 = p.zb + p.Lb1 * sin(p.alpha_b);
    xs = p.xb + p.Lb * cos(p.alpha_b);
    zs = p.zb + p.Lb * sin(p.alpha_b);

    xB = x3 + d4 * cos(q3);
    zB = z3 + d4 * sin(q3);

    L_h = sqrt((xB - xs)^2 + (zB - zs)^2);

    Lb2 = p.Lb - p.Lb1;
    h_h = (d4 * Lb2 * sin(p.alpha_b - q3)) / L_h;
end

function v_w = compute_3d_velocity(th1, q3, d4, th1_d, q3_d, d4_d, p)
    rT = d4 + p.Ltx;
    J11 = -rT*sin(q3) + p.Lty*cos(q3);
    J12 = cos(q3);
    J21 = rT*cos(q3) + p.Lty*sin(q3);
    J22 = sin(q3);

    xT0_d = J11 * q3_d + J12 * d4_d;
    zT0_d = J21 * q3_d + J22 * d4_d;

    x3 = p.xb + p.Lb1 * cos(p.alpha_b);
```

```
    xT0 = x3 + rT * cos(q3) + p.Lty * sin(q3);

    v_w = [xT0_d * cos(th1) - xT0 * th1_d * sin(th1);
           xT0_d * sin(th1) + xT0 * th1_d * cos(th1);
           zT0_d];
end
```

6.9 Research Supervisor Summary & Master Defense Checklist

Core Concepts Mastered in Chapter 6

1. **Topological Kinematic Tree:**

The electric rope shovel is formally modeled as a kinematic tree consisting of 5 rigid link subsystems and 4 joints, closed by the parallel **Hoist Cable Constraint Loop** C_h.

1. **Unified Reference Frame Hierarchy:**

Orthogonal coordinate frames $\{W\} \to \{0\} \to \{1\} \to \{2\} \to \{3\} \to \{4\} \to \{5\} \to \{T\}$ map 3D positions unambiguously without gimbal lock representation singularities.

1. **Closed Hoist Loop Geometry:**

The suspended cable length $L_h(q_3, d_4)$ and hoist moment arm $h_h(q_3, d_4)$ are derived in strict closed form:

$$h_h(q_3, d_4) = \frac{d_4 L_{b2} \sin(\alpha_b - q_3)}{L_h(q_3, d_4)}$$

The lifting mechanical advantage reaches its peak when the dipper handle is perpendicular to the boom chords ($\alpha_b - q_3 \approx 90^\circ$).

1. **Industrial Benchmark Reconciliation:**

Full parameter provenance for both the $20\,\mathrm{t}$ P&H 2100BL benchmark and the $55\,\mathrm{t}$ TZ WK-20 deployment shovel is established and cross-verified against technical assembly drawings.

Doctoral Viva & Defense Questions

- **Q1:** *Why is the hoist moment arm h_h a function of both saddle angle q_3 and crowd extension d_4, rather than q_3 alone?*

Answer: As the crowd extends ($d_4 \uparrow$), the dipper bail pin B moves farther from the saddle pivot O_3. This changes the triangle formed by O_3, the sheave center S, and the bail pin B. While the angle $(\alpha_b - q_3)$ depends purely on q_3, the suspended rope length L_h in the denominator varies strongly with d_4, altering the direction of the cable vector and shifting the line-of-action moment arm.

- **Q2:** *Why does the swing degree of freedom θ_1 not affect the vertical dynamics $\ddot{z}_T$?*

Answer: The slewing axis Z_0 is collinear with the gravity vector $\mathbf{g} = [0, 0, -g]^T$. Rotation about a vertical axis is strictly in the horizontal plane (X_0-Y_0). Consequently, the vertical position $z_T^W = Z_c + z_T^0(q_3, d_4)$ has zero partial derivative with respect to θ_1, decoupling vertical gravity forces from the swing angle.

- **Q3:** *Under high-speed slewing, what spatial inertial force acts on the dipper handle?*

Answer: Slewing at angular rate $\dot{\theta}_1$ creates two major spatial inertial forces: a radial centrifugal force flinging the entire front-end outward ($F_{\text{cent}} = M_d x_T^0 \dot{\theta}_1^2$), and a lateral Coriolis force ($F_{\text{Coriolis}} = 2M_d \dot{\theta}_1 \dot{x}_T^0$) acting perpendicular to the boom plane. This lateral force produces severe out-of-plane torsional twisting on the boom foot pins and shipper shaft bushings.

Transition to Chapter 7

With the complete 3D kinematic topology, reference frames, and cable loop geometry formulated, we proceed to:

Volume II, Chapter 7: Planar 2-DOF Crowd-Hoist Dynamics and Cable-Driven Kinematics.

In Chapter 7, we formulate the complete dynamic equations of motion including dynamic hoist rope elasticity, sheave rotational inertia, drum winding dynamics, and motor drive transmissions.